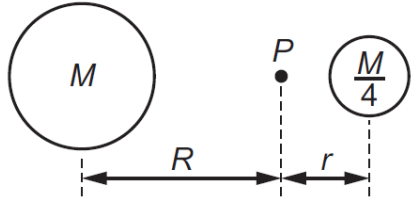
10 Two large masses, one of mass M, the other of mass $\frac{M}{4}$, are positioned as shown.



A small mass is placed at point P such that it experiences zero gravitational force from the masses.

What is the ratio $\frac{R}{r}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4